

BIT1123 OBJECT-ORIENTED PROGRAMMING

ASSIGNMENT 1 – SELF-REFLECTIVE REPORT

Student Information

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Programme: BSSCE

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1. Introduction

This report is a reflection of the learning experience I had during Object Oriented Programming tutorials 1 – 10. The tutorials covered primarily Java programming and introduced the various concepts in Object Oriented Programming in a practical way. The tutorials helped me learn the fundamentals of java programs and their structure and syntax at the start. The weeks went on and I started using classes, objects, inheritance, polymorphism, abstraction and graphical user interfaces. The practical activities helped me understand the structure of the java programs, how OOP concepts can be applied in solving programming problems. I also got some experience in structuring my programming with GitHub during the tutorials. I learned that it would be easier to manage and review programming projects if they were kept organized in separate folders containing the source code.

2. Knowledge Gained from Each Tutorial

Week 1

In Week 1, I learned the basic structure of Java programs and how to write and run simple Java code. I worked with programs such as HelloWorld.java and StudentGrade.java.

This tutorial helped me understand basic Java syntax, variables, output statements, and simple programming logic. It provided a foundation for the more advanced topics covered later.

Week 2

In Week 2, I was introduced to classes and objects. I worked with Student.java and Main.java.

I learned that a class can be used as a blueprint for creating objects and that objects can contain attributes and methods. This was an important step in understanding the main idea behind Object-Oriented Programming.

Week 3–4

During Weeks 3 and 4, I worked with classes such as Person, Student, and Lecturer. These exercises helped me understand relationships between classes and the concept of inheritance.

I learned how a child class can inherit properties and methods from a parent class. This helped me understand how inheritance can reduce repeated code and make programs more organized.

Week 5

Week 5 gave me more practice with Object-Oriented Programming using Student.java and Main.java.

I became more comfortable creating classes, defining methods, creating objects, and organizing Java programs into separate files. This tutorial also helped strengthen the concepts that I learned in the earlier weeks.

Week 6

In Week 6, I worked with Employee.java, Lecturer.java, and Main.java.

The main learning point was working with inheritance and relationships between classes. I gained a better understanding of how a subclass can extend a parent class and use its existing features.

Week 7

Week 7 introduced more advanced OOP concepts through classes related to appliances, including Microwave and WashingMachine.

This tutorial helped me understand abstraction and polymorphism. I learned that different objects can share common characteristics while still having their own specific behaviour.

Week 8–9

Weeks 8 and 9 provided additional Java programming practice. These activities allowed me to apply the concepts learned in previous tutorials and improve my ability to write and test Java programs.

I also became more comfortable identifying errors and making changes to my code when the expected output was not produced.

Week 10

Week 10 introduced Java GUI programming. I worked with Questions.java and QuizBattleGUI.java to create a simple quiz application.

This was one of the more interesting tutorials because I was able to see how Java could be used to create an application with a graphical interface. I learned about GUI components and how user actions can be handled within a Java program.

3. Challenges Encountered

This was one of the most difficult things I had to learn during the tutorials: to know how to program in Java and how to program in Object Oriented Programming.

Understanding inheritance, abstraction and polymorphism was hard because they require us to consider relationships between multiple classes. Another problem was one of detection and correction of errors in the java programs. A minor syntax error or wrong class association may cause the program to be executed incorrectly. The reason for an error would have to be examined, and it would be necessary to review my code and run it once again. The GUI tutorial was also difficult due to the introduction of new concepts in the GUI aspects and in user interaction.

4. How the Challenges Were Overcome

I managed to get around these problems by practising the examples given in the tutorials, and by running the programs a number of times. When I was looking at the final result, I attempted to comprehend the operation of each class, method and object. If any mistakes occurred, I made sure to carefully read my code and fix the error before rerunning. In addition, I used Java compiler and development tools to detect errors. With the OOP concepts, I learned about the relationship between classes and objects through practicing with various examples. These ideas were repeated over the course of the tutorials, which helped them to grasp them over time.

5. Improvements in Java Programming Skills

The tutorials were quite useful in improving my Java programming abilities. Initially, I learned simple Java syntax and programming statements, and this was my main understanding. I felt more comfortable with making more classes and linking them up after going through the tutorials. I also learned to design objects, implement methods and understand inheritance, and organized Java source files correctly. Learned to read error messages and diagnose the problem in my programs rather than rewrite the code. I also gained some knowledge of the organization of a Java project through the tutorials. Keeping the work from the tutorial in a repository on GitHub helped me appreciate the need for structure and accessibility of programming projects.

6. Understanding of Object-Oriented Programming Concepts

The tutorials helped me gain more insight into the fundamental concepts of Object Oriented Programming. Classes and Objects: I learnt that a class is a blueprint and an object is an instance of a class. Encapsulation: I learned that data and methods can be bundled in a class and controlled by appropriate access. Reusing code: I learned that a child class can inherit attributes and methods from a parent class. Polymorphism: I learned that objects from different classes can be used using a common parent type but have different behaviour. Abstraction: Implemented details can be hidden and only important features shown to user. By grasping these concepts, I realized the potential of OOP in structuring programs, reusability, and maintainability.

7. Future Learning Plans

I would like to build larger applications rather than just complete small tutorial exercises to enhance my Java programming skills in the future. I would like to be more confident with advanced OOP concepts and how they're applied in the real software projects. I will also be working on enhancing my knowledge of the Java GUI development and will try to learn some other java technologies. Furthermore, I would like to continue using GitHub for project management and develop my programming portfolio. Another is to enhance my debugging and problem-solving skills which are key when writing real-world software.

8. Conclusion

The Object Oriented Programming tutorials (1-10) were good to get a solid understanding of Java and OOP. I began programming from scratch in Java and gradually learned how to write classes and create objects and use concepts like inheritance, abstraction and polymorphism. I had some problems while learning new OOP concepts and debugging some errors, but regular practice allowed me to get better. The tutorials also provided me a practical experience in organizing my source code and using GitHub as a repository for my source code. This assignment helped me to review how far I have come and to see the improvements I've made in my programming skills with java. I think that I'll use the knowledge I learned in these tutorials in the future for programming and projects.

9. GitHub Repository URL

GitHub Repository:

https://github.com/awab1204643-coder/awabali_202305010141_OOP_Tutorials